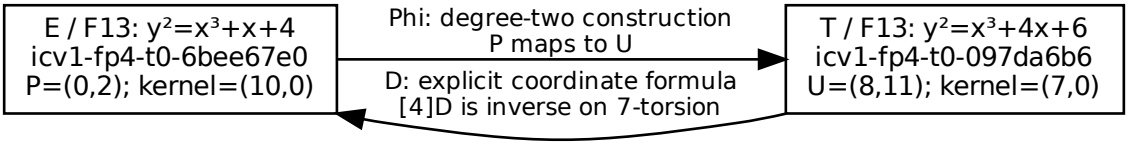
Explicit F13 rational-point dual and subgroup inverse

2026-10-29; PR2381; cm8-dual-20261029

Verified finite result. Independent C++ and Rust arithmetic check all14 source and14 target rational points,196 addition pairs in each direction, both compositions equal to doubling, and all7 odd-subgroup inverses. Seven valid target points not killed by7 are excluded from the inverse domain. No invalid-curve experiment or generic geometric-duality claim is made.



The historical identity bindings above are missing from the merged active registry; this does not install or validate a combined catalog. Both curves are nonsingular. Put $v = 12x - 6$ on T. The new actual point function is

$$D(x,y) = \left(10(v + 5 + 8/v) + 10,\ 12y(1 - 8/v^2)\right), \qquad I(R) = [4]D(R).$$

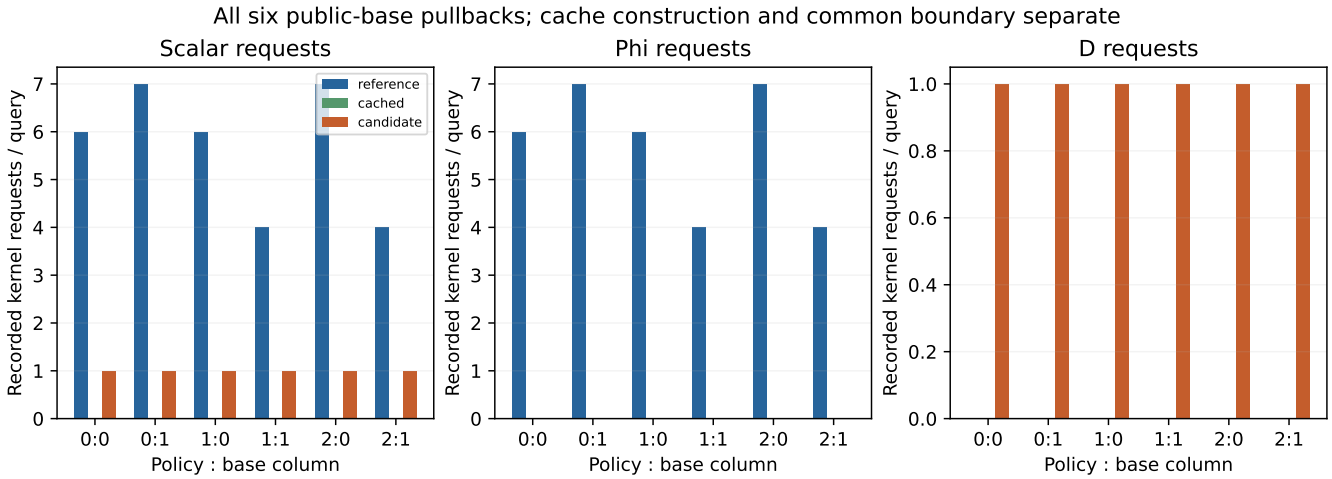
Infinity and the rational kernel $(7, 0)$ map to infinity. The forward map is $\Phi(x,y) = (12(x + 2/(x - 10)), 8y(1 - 2/(x - 10)^2))$, with source kernel $(10, 0)$. Native and kernel-checked finite results give $D\Phi = [2]$, $\Phi D = [2]$ and $I\Phi(Q) = Q$, $\Phi I(R) = R$ when the respective point is killed by7. This removes enumeration from the specified public-base inverse computation; public fixture/base construction and the rest of setup remain separate.

Full S3 preservation. Both native attempts execute the frozen12 arms and 72 separate nonidentity main-source recoveries. Every arm/index byte equals the reference and original CM8 record: roots, witnesses, ordered signs/lifts, useful rows/ranks, original/reduced degree, setup logs and source scalars. The unchanged independent Rust S3 auditor checks both streams. All nine altered, rehashed controls reject101 with no receipt;22 backend tests pass. The two runs differ only in source provenance and a corrected receipt-key label; all26 reference/candidate arm/index files match exactly, and all non-source fields of dual.json match. Earlier mislabelled receipt is retained.

Formal scope. New CM8Dual.lean has21 named theorems /25 audited declarations, zero failing or extra trusted/native axioms under the unmodified scoped gate. Only propext, Classical.choice and Quot.sound occur. Actual Mathlib E/T point maps use coordinate formulas, not logarithm tables. Exhaustive coordinate coverage proves that the finite proof tables include every rational point; then both full addition/composition statements follow. A general additive-map lemma explicitly assumes $D(FQ) = 2Q$, $2c = r + 1$ and $rQ = 0$, and proves $cD(FQ) = Q$. The source-scalar theorem exposes annihilators for both unknown Q and generator P. Its actual F13 instance accepts arbitrary natural scalars, not only supplied generator-index labels. Geometric duality over extensions or arbitrary fields is not inferred from finite F13 points.

Lean4.34.1, Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. The first proof build failed at the final coercion simplification; exact source and logs are retained. An explicit reduction fixes the same statement. Old projects and17 polynomial certificates remain unchanged. The existing universal translated-construction IDC1h45f7f67f60c827c3 is replayed; it certifies polynomial arithmetic, not the whole dual/group claim. The new finite point proofs are separate. No fresh combined count over historical projects is claimed.

Request comparison, controls and remaining objective



Recorded requests. The declared cached reference builds seven public images once (7scalar and7Phi requests), then performs one lookup per query. Each route additionally executes the same boundary:2scalar7 checks,1Phi verification,2curve checks,3group comparisons. These are differently weighted and sometimes nested requests, not time or total cost. The declared cache is not a strongest optimized enumeration implementation. Field arithmetic, internal scalar work, memory, lookup engineering and audit remain unpriced.

Target base point	D(target)	[4]D(target) = reference inverse
$(6, 5)$	$(7, 9)$	$(9, 1)$
$(8, 2)$	$(9, 1)$	$(0, 11)$
$(11, 4)$	$(0, 11)$	$(7, 9)$

Each appears twice in the six policy/column entries. The reference kernel uses 6,7,6,4,7,4 scalar/Phi/comparison requests respectively; the direct kernel uses one D and one scalar4 request each. These counts do not establish a speed ratio.

Yield and degree unchanged. The earlier fixed-target CM8 comparison keeps 3of54 reduced joint gains and0original, with all four matched degree differences favoring the source. The preceding uniform96arm/896target/210policy study keeps 0joint gains in both input forms. No new useful-yield, intrinsic regularity, F4/F5 or calibrated complete-work advantage is claimed by an inverse change. The original objective remains open.

Execution flow. One fixed native contract, two executed full attempts, both completed, none interrupted/excluded/resumed. Attempt02 follows a frozen metadata correction: "nontorsion" was a misleading receipt key because all finite group points are torsion; the correct category is not killed by7. The old successful source/output remains. The failed compiler adapter and proof build are preparation/formal failures, not hidden successful runs. Each correction was committed before affected replay; every source hash is bound. The exact legacy Markdown protocol was committed before data; a later structured export is metadata, not a retroactively preregistered JSON protocol or canonical protocol-checker approval.

Validation and holds. Native full replay, nine corruptions, existing certificate17, scoped Lean gate and backend tests pass. Root library validation is separately recorded and must not be inferred green. Active1921-model registration/catalog, naming, full library, Conductor (127/no lease), exact-head CI/review/integration and broader completion gates remain. No paid resource or arbitrary assets upload. Approved official branch checkpoints succeeded. Native/proof caps600sec, science envelope18:47–19:47UTC, two local workers; no native cap hit. No wall-time, GAE/S/rho/external-fastest/leaderboard promotion.

Sources and reproduction. Frozen CM8 forward source/independent audit, CM8Transfer.lean, explicit-dual card and new preregistered protocol; known isogeny-transfer/presentation-invariance atlas leads, no invention claim. Run `run-cm8-dual.sh` with a fresh outside directory after the official current checkpoint gate. Native records, complete raw restore, proofs and source versions are linked from the study README. All older frozen evidence and both catalog provenance anchors remain. Next work needs actual useful-yield/ relevant-degree evidence on a surviving encoding, plus complete main-DLP work; a verified inverse alone does not supply it.